



**DEPARTMENT OF COMPUTATIONAL INTELLIGENCE
III YEAR CSE - AIML II SEM**

COURSE: APPLICATION DEVELOPMENT - 2

COURSE CODE: R22A66933

**ADVANCED LANGUAGE DETECTION USING CONVOLUTIONAL NEURAL
NETWORKS FOR ACCURATE MULTILINGUAL TEXT ANALYSIS**

ABSTRACT

Language detection is a critical task that allows systems to identify the language of a given text automatically. This capability is essential for applications such as multilingual content processing and machine translation. Existing systems primarily use statistical models or rule-based approaches to detect languages, relying on character-level or word-level patterns. While effective for common languages, these methods struggle with short or ambiguous texts. The key drawback is limited accuracy when dealing with noisy or brief text inputs. Our work proposes an enhanced language detection framework using Convolutional Neural Networks (CNNs) to improve detection accuracy, particularly for short and ambiguous texts. The model leverages CNN's ability to capture local patterns and features in the text for better generalization across languages. Technologies include the implementation of CNNs with libraries like TensorFlow and Keras, alongside data pre-processing techniques for effective text classification.

Keywords: Language Detection, CNN, Deep Learning, Text Classification

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